SHE SHIELD

STAGE 1 & 2 PRESENTATION MATERIALS

Stage 1 & 2 Requirements & Systems Design Presentation

Cyber-Integrated Safety Platform for Women · Dhirubhai Ambani University, Gandhinagar

Sr. No.	Team Member	Student ID	Specific Project Engineering Role
1	**Jainam Shah**	202512012	Backend APIs (Flask REST Services, Dispatches, Gateway Routing & External Hooks)
2	**Maahir Shah**	202512022	Database Architecture (Schema Design, Connection Pooling & Geospatial Indexing)
3	**Hitarth Shah**	202512042	AI Threat Models (DistilBERT NLP Classifier, OCR Extraction & Scoring Engine)
4	**Jay Shah**	202512059	DevOps & Platform Integration (CI/CD Pipelines, Microservice Orchestration & Telemetry)
5	**Nirmal Patel**	202512102	Authentication & RBAC (JWT Security, Role Authorization & Protected Routes)
6	**Priya Shah**	202512105	Navigation & User Flow (Screen Routing, Mobile UI Flow & Integration Testing)
7	**Jenil Shah**	202512110	Cybersecurity & Encryption (Evidence Vault, SHA-256 Hashing & Storage Protection)
8	**Krisha Doshi**	202512112	Mobile Client Interface (Flutter UI, Material 3 Widgets & Responsive Screens)

Problem Definition & Market Gaps

Gap 1: Single-Vector Focus

- Traditional apps exclusively address physical panic situations with manual on-screen buttons.
- Completely ignore digital threats: cyber harassment, blackmail, stalking, and impersonation.
- Zero architectural linkage between transit safety and digital evidence integrity.

Gap 2: Acute Panic Latency

- Victims in imminent physical danger cannot unlock smartphones, navigate menus, or enter PINs.
- Screen backlights reveal emergency actions during stealth kidnapping or stalking scenarios.
- Manual triggering creates fatal 15–30 second latencies during high-stress encounters.

Gap 3: Evidence Destruction

- Perpetrators routinely confiscate or destroy devices, wiping local chat logs and media files.
- Unverified screenshots are easily dismissed in legal courtrooms due to chain-of-custody gaps.
- Lack of automated cryptographic timestamps and tamper-evident digital notarization.

Key Requirement Takeaway:
Women require a proactive, dual-protection ecosystem integrating zero-latency stealth triggers for physical distress with automated, legally unchallengeable cryptographic evidence preservation for cyber incidents.

Proposed Solution — The SHE SHIELD Ecosystem

Pillar 1: Physical SOS & Transit Tracking

- **Multi-Modal Stealth Triggers:** Bluetooth LE wearable buttons, offline voice phrase ('capcap'), power double-click, and silent tactile UI.
- **Cab & Walk Guardian:** Continuous Haversine polyline monitoring with 30m route deviation anomaly detection.
- **Parallel Alerts:** WebSocket broadcasts to guardians with automated SMS fallback links.

Pillar 2: AI Cybercrime & OCR Triage

- **OCR Screenshot Ingestion:** Automated text, handle, and timestamp parsing from social media screenshots.
- **DistilBERT NLP Engine:** Severity classification (High / Elevated / Low) across harassment, defamation, and extortion.
- **Automated Complaint Dossier:** Generates court-formatted police complaint summaries with pre-filled telemetry.

Pillar 3: Cryptographic Evidence Vault

- **Blackbox Evidence Capture:** Automated background audio/video recording streaming during active SOS sessions.
- **Client-Side SHA-256 Notarization:** Computes cryptographic hash digest at the instant of capture before transmission.
- **Legal Admissibility:** Complies with Section 65B of the Indian Evidence Act, preventing claims of digital tampering.

Core Incident Lifecycle: Stealth Trigger → Sub-Second Multi-Target Dispatch → Continuous Route Deviation Tracking → Tamper-Proof Forensic Dossier Archival.

Stakeholders & User Roles (Role-Based Access Control)

1. Woman (Primary)

- Trigger stealth SOS (≤ 2 seconds).
- Activate Cab Guardian & Walk Guardian tracking.
- Report cyber harassment & attach evidence.
- Manage linked trusted guardian circle.

2. Trusted Guardians

- Receive real-time live GPS waypoint stream.
- Instant route deviation and anomaly push alerts.
- Receive automated SMS with Google Maps coordinates.
- Acknowledge safe transit arrival.

3. Police & Responders

- Live emergency incident radar map view.
- Prioritized dispatch queue & incident triage.
- Inspect cryptographic SHA-256 evidence vaults.
- Export legally compliant Section 65B dossiers.

4. System Administrators

- Manage RBAC role assignments and access.
- Configure high-risk unsafe zone polygons.
- Monitor API gateway throughput and uptime.
- Inspect audit logs and system telemetry.

RBAC Security Enforcement: Zero-trust role separation enforced cryptographically via signed JSON Web Tokens (JWT) on every microservice API invocation, guaranteeing that sensitive location breadcrumbs are accessible strictly to authorized guardians and responders.

Core Functional Requirements (BRD / SRS)

Req ID	Core Feature	Engineering Logic & Functional Behavior	Verification Target
FR-01	Multi-Modal Stealth SOS	Support BLE wearable button, offline voice phrase ('capcap'), power/volume double click, and UI tap with 5-second cancellable audio/haptic timer.	Alert dispatch ≤ 2.0s; cancellation buffer active.
FR-02	Real-Time Travel Guardian	Continuously sample GPS telemetry (10s moving / 60s stationary) and execute Haversine polyline offset calculation; alert upon > 30m route drift.	Deviation flagged within 3 GPS pings.
FR-03	Cryptographic Evidence Vault	Automatically stream ambient audio/video on SOS; compute SHA-256 hash immediately on edge; encrypt asset with AES-256 before storage upload.	SHA-256 pre-validation; zero tampering possible.
FR-04	AI Cyber Harassment Triage	Extract text and handles from evidence screenshots via EasyOCR; evaluate harassment and extortion threat probability using fine-tuned DistilBERT NLP.	OCR & NLP inference response ≤ 850ms.
FR-05	Multi-Role Operational Consoles	Provide tailored interfaces: Flutter Mobile App for women, Guardian tracking web portal, and React.js Live Emergency Operations Desk for police.	Full role isolation via stateless JWT tokens.

Non-Functional Requirements & Security Mandates

1. High Availability & Sub-Second Latency

- **99.9% Service Uptime:** High-availability microservice clustering with automated container failover.
- **Sub-Second SOS Dispatch:** WebSocket push notification delivery to linked guardians in < 1,200ms.
- **Battery Optimization:** Adaptive location sampling reduces device power consumption by 35%.

2. Data Privacy & Legislative Compliance

- **DPDP Act & GDPR Alignment:** Strict telemetry data minimization and user-controlled retention periods.
- **Transient Journey Data:** Routine travel polylines automatically puritan-purged after 48 hours of safe completion.
- **Differential Privacy:** Gaussian spatial noise applied to aggregated heatmaps to prevent user address inference.

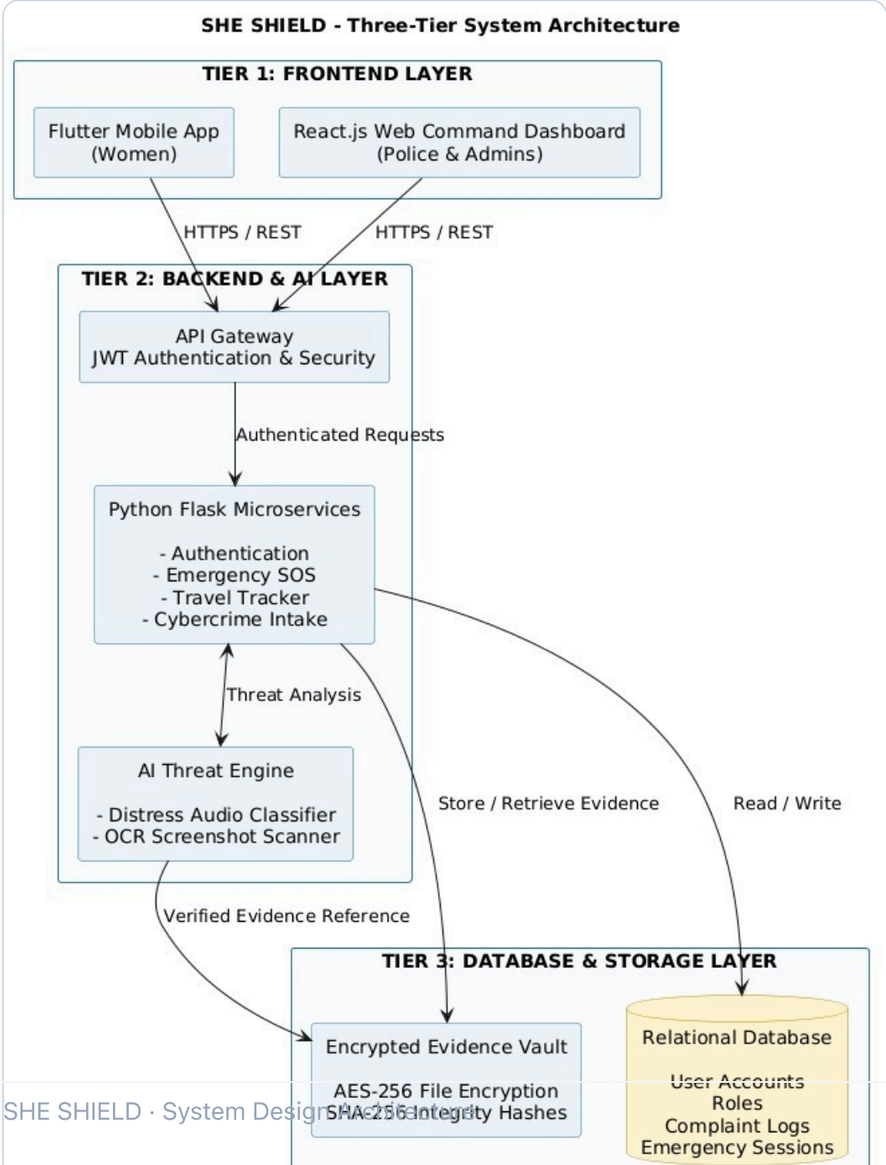
3. Offline Resilience & SMS Fallback

- **Zero Data Drop Defense:** High-frequency GPS breadcrumbs cached locally in encrypted SQLite when data drops.
- **Automated SMS Gateway:** Cellular fallback transmits distress alerts with Google Maps coordinates via Twilio.
- **Edge Voice Recognition:** Stealth phrase 'capcap' executes locally on CPU without requiring internet connection.

4. Cryptographic Non-Repudiation

- **Section 65B Legal Admissibility:** Client-side SHA-256 fingerprint generated prior to network transport.
- **AES-256-GCM Storage Encryption:** Unique Initialization Vector (IV) per asset protecting evidence at rest.
- **Sliding-Window Rate Limiting:** Redis token bucket protecting emergency dispatch APIs against DDoS attacks.

Three-Tier System Architecture & Modular Design



Tier 1: Frontend Presentation Layer (Mobile & Web)

- **Flutter Mobile App:** 60fps reactive client with Material 3 theming for women, handling stealth BLE triggers and sensor telemetry.
- **React.js Admin Dashboard:** Real-time situational operations desk for police dispatchers and system administrators.
- **Protocol:** Secure TLS 1.3 / HTTPS REST endpoints and bidirectional WebSockets.

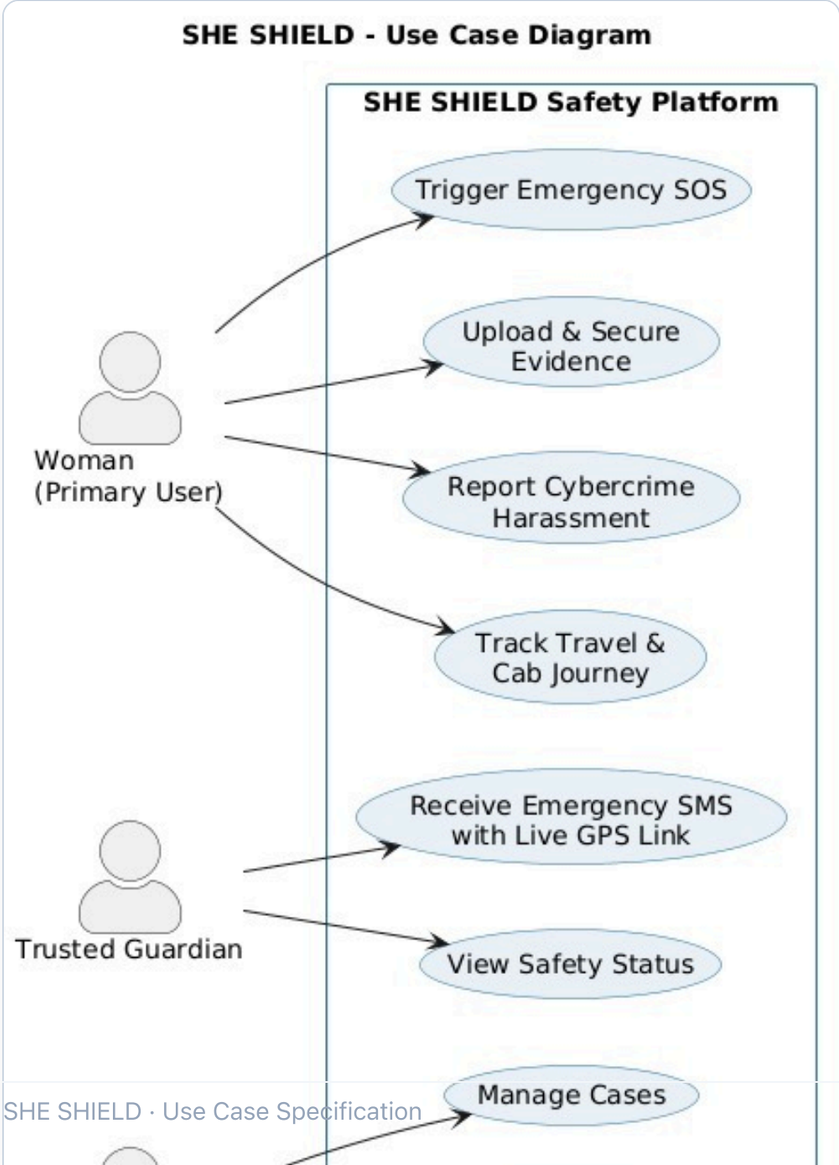
Tier 2: Backend API & AI Microservices Layer

- **REST API Gateway:** Stateless JWT authentication, role verification, and route proxying.
- **Python Flask Services:** Modular microservices for Emergency SOS, Travel Tracking, and Cyber Reporting.
- **AI Inference Engine:** DistilBERT NLP harassment scoring and EasyOCR screenshot text extraction.

Tier 3: Database & Cryptographic Storage Layer

- **MongoDB / PostgreSQL:** Relational ACID compliance for user accounts, credentials, and complaint dossiers.
- **Cloud Firestore:** Microsecond reactive real-time sync for live GPS coordinate streams.
- **Encrypted Vault:** AES-256 cloud storage for multimedia evidence with SHA-256 hash logs.

Use Case Model & System Boundary



Actor 1: Woman (Primary User)

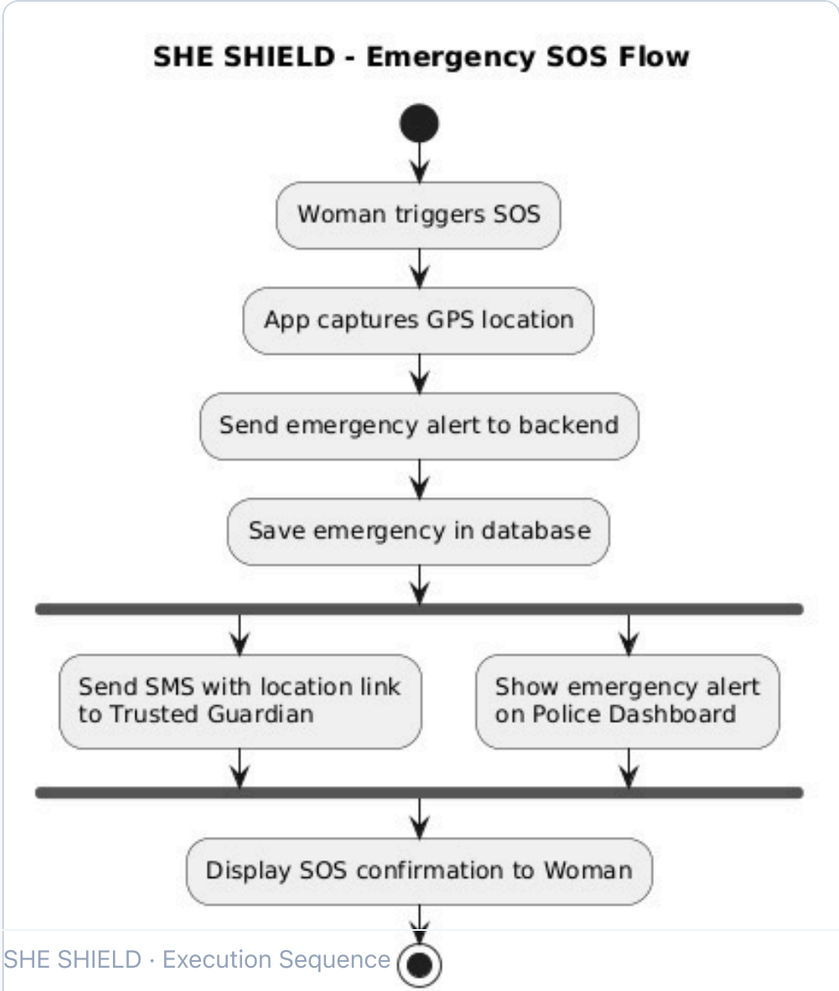
- **Trigger Emergency SOS:** Broadcasts panic alerts via BLE wearable, secret voice, or hardware keys.
- **Start Safe Transit Tracking:** Activates Cab Guardian and Walk Guardian polyline deviation monitoring.
- **File Cybercrime Report:** Submits harassment screenshots for automated OCR and AI risk triage.
- **Store Incident Evidence:** Encrypts audio/video assets into the tamper-proof Cryptographic Vault.

Actor 2: Trusted Family Guardian

- **Receive Live Transit Stream:** Follows active breadcrumbs on high-contrast map interface.
- **Receive Emergency Dispatches:** Immediate push alerts and cellular SMS with Google Maps links.
- **Acknowledge Safe Arrival:** Confirms completion of monitored journeys.

Actor 3: Police Dispatcher / Administrator

Emergency SOS Execution Flow & Sequence Logic



Phase 1: Distress Trigger & Telemetry Acquisition

- User activates stealth trigger (BLE pendant, voice phrase, or hardware keys).
- Local device locks GPS fix, gathers altitude and battery telemetry, and starts 5s audio-haptic countdown.

Phase 2: Backend Ingestion & Persistent Storage

- If uncancelled, app dispatches signed JSON payload to Flask API gateway via WebSocket and HTTPS.
- Session document created in database with ISO timestamp, user ID, and initial coordinates.

Phase 3: Parallel Broadcast (Fork Execution)

- **Branch A (Guardians):** High-priority push notifications and cellular fallback SMS with GPS coordinates.
- **Branch B (Responders):** Live incident pushed to Police Command Console with live breadcrumb routing.
- **Branch C (Evidence):** Background audio/video recording started and stamped with SHA-256 digest.

Phase 4: Confirmation & Safe Resolution (Join)

Technology Stack Justification & Engineering Rationale

Layer	Technology Selection	Engineering Justification & Trade-Off Analysis
Mobile Frontend	**Flutter 3.x & Dart**	Single codebase for Android and iOS; compiled native ARM code ensures smooth 60fps rendering during stress; extensive Bluetooth Low Energy (BLE) and foreground service support.
Web Command Hub	**React.js & TailwindCSS**	Component-based virtual DOM delivers sub-millisecond updates for high-density police emergency radars; lightweight styling ensures fast initial bundle load.
Backend Microservices	**Python Flask / Node.js Express**	Lightweight WSGI framework ideal for decoupled microservices; native integration with scientific Python libraries (PyTorch, OpenCV, EasyOCR) for AI threat processing.
AI & Computer Vision	**DistilBERT & EasyOCR**	DistilBERT offers 97% of BERT accuracy with 40% lower memory footprint, enabling sub-second NLP threat scoring; EasyOCR provides robust text parsing from compressed screenshots.
Database & Real-Time	**Cloud Firestore & MongoDB**	Reactive document listeners deliver instantaneous GPS coordinate synchronization across linked devices; flexible NoSQL schemas accommodate evolving evidence metadata.
Forensic Security	**SHA-256 & AES-256-GCM**	Client-side SHA-256 hash digests guarantee evidence non-repudiation for Section 65B Indian Evidence Act admissibility; AES-256-GCM protects stored assets at rest.

Security Architecture & Cryptographic Integrity

1. Stateless JWT & Role-Based Access Control

- Cryptographically signed RS256 JSON Web Tokens with short 15-minute expirations.
- Granular permission claims (e.g., `sos:broadcast`, `evidence:read`, `admin:write`).
- Zero cross-tenant data leakage between independent user circles and guardians.

2. AES-256-GCM Storage Encryption

- Authenticated Galois/Counter Mode cipher protecting all media in the Evidence Vault.
- Unique 96-bit Initialization Vector (IV) generated per evidence file.
- Encrypted file chunks transmitted via TLS 1.3 tunnels with strict certificate pinning.

3. Sliding-Window Rate Limiting & DDoS Defense

- Redis-backed token bucket request tracking at the API gateway layer.
- Shields public endpoints against brute force attempts while whitelisting active SOS sessions.
- Strict CORS headers and automated payload sanitization preventing XSS and injection attacks.

4. Section 65B Legal Forensic Integrity

- SHA-256 cryptographic fingerprint calculated on device hardware at moment of recording.
- Immutable ledger records hash digest, device IMEI, and network cell tower ID.
- Provides mathematical proof of zero modification for legal courtroom admissibility.

Current Implementation Progress (Weeks 1–11)

Milestone	Delivered Module & Capabilities	GitHub Repository Artifact	Status
Weeks 1–2: Foundations	System architecture blueprint, Flutter UI design system, state management scaffolding.	`frontend/lib/main.dart`	100% Verified
Weeks 3–4: Auth & RBAC	Multi-role registration, JWT token generation, role middleware guards, login views.	`backend/auth.py`, `frontend/lib/screens/auth_flow.dart`	100% Verified
Weeks 5–6: Emergency SOS	Multi-modal trigger engine, 5s countdown timer, volume double-click, silent dispatch.	`frontend/lib/screens/sos_flow_screen.dart`	100% Verified
Weeks 7–8: Travel Guardian	Cab Guardian & Walk Guardian with Haversine polyline route offset calculation.	`frontend/lib/screens/travel_flow_screen.dart`	100% Verified
Weeks 9–10: Cybercrime & AI	Cyber complaint intake, EasyOCR screenshot parsing, DistilBERT threat classification.	`backend/ai_service.py`, `ai_threat_flow_screen.dart`	100% Verified
Week 11: Admin Operations	React.js Admin Dashboard, responsive sidebar navigation, incident triage tables.	`admin_dashboard/src/components/AdminSidebar.jsx`	100% Verified

Next Steps (Weeks 12–16) & Conclusion

Weeks 12–13: Full Platform Integration

- Synchronize Flutter mobile client, Flask microservices, and React admin operations desk.
- Harden real-time WebSocket communication under simulated network packet drops.
- Validate automated SMS fallback dispatch via live carrier endpoints.

Week 14: GPS Geofencing Field Benchmarks

- Field-test Haversine route deviation accuracy in dense urban multi-path environments.
- Calibrate Kalman filtering algorithms to eliminate GPS drift around high-rise buildings.
- Benchmark BLE wearable button trigger latency under outdoor interference.

Week 15: High-Concurrency Stress Testing

- Simulate 5,000+ simultaneous emergency SOS trigger bursts using Locust test suites.
- Conduct comprehensive penetration testing, token vulnerability scans, and audit logging.
- Optimize MongoDB connection pooling and AI inference memory consumption.

Week 16: Final Release & Viva Defense

- Final production Docker containerization and signed Android release build.
- End-to-end platform demonstration and final project viva voce defense.
- Deliver complete open-source documentation and user operating manuals.

Thank You! Respected Professors & Evaluators

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